Big Data Cup 2026 (Rough Draft v1)

Off-Puck Space Creation Value

Working Draft

February 2026

Question. Which players and teams create offensive value *without* puck touches by moving defenders, opening seams, and increasing shot opportunity quality?

Data + scope. 10 games from the public BDC 2026 release, 5v5 only. We segment 4,257 offensive-zone possessions and evaluate 83,149 off-puck frame contributions (frame stride = 25).

1. Approach Summary

For each sampled frame in an offensive possession:

- Build an **Opportunity Field** from (i) slot openness, (ii) seam openness, and (iii) shooter release space.
- Attribute each non-carrier skater's impact with a leave-one-out freeze:

$$\Delta O_t^{(A)} = O_t(\text{all}) - O_t(\text{player A frozen at prior position}).$$

- Define **Space Creation Added (SCA)** as summed/averaged ΔO_t over off-puck windows.
- Detect off-puck action tags (cut, drag, screen, decoy) with rule-based motion/location logic.

Validation targets. Primary target: possession ends in shot/goal. Secondary: slot shot/goal.

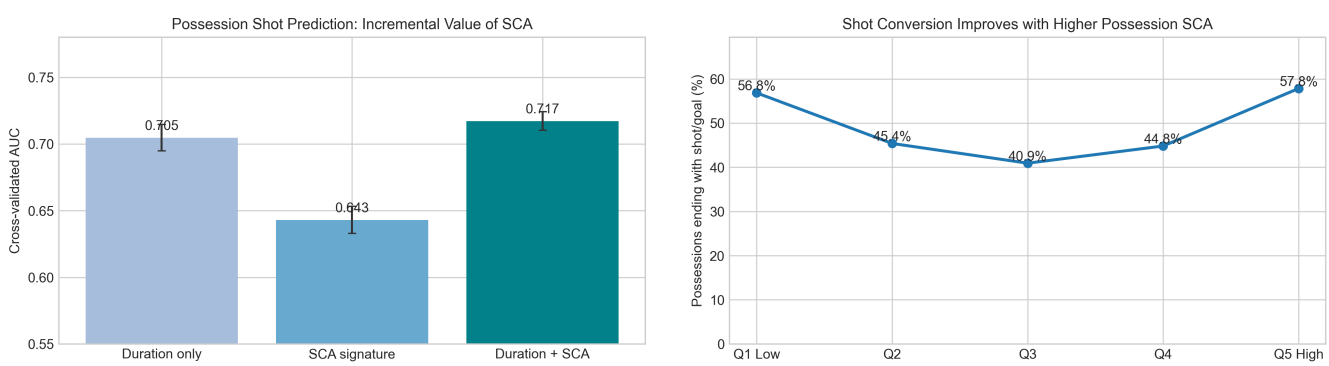

Figure 1: Left: SCA adds signal on top of possession duration for shot prediction. Right: shot rates by possession SCA quintile (descriptive, unadjusted).

2. Core Findings

SCA adds incremental predictive value.

- Duration-only model: AUC $= 0.705 \pm 0.010$.
- SCA-only signature: AUC $= 0.643 \pm 0.010$.

- Duration + SCA: AUC $= 0.717 \pm 0.007$.

Interpretation: off-puck features are not a replacement for possession length/context, but they provide additive signal.

Archetypes emerge from movement style. Clustering identifies at least four robust archetypes: low-activity fillers, drag/cut seam manipulators, screen-dominant net-front players, and balanced support movers.

Action-level effect estimates. In this baseline model, drag and cut actions show positive mean lift on opportunity, while screen/decoy flags are negative on average (likely because many occur in already-crowded late-phase possessions).

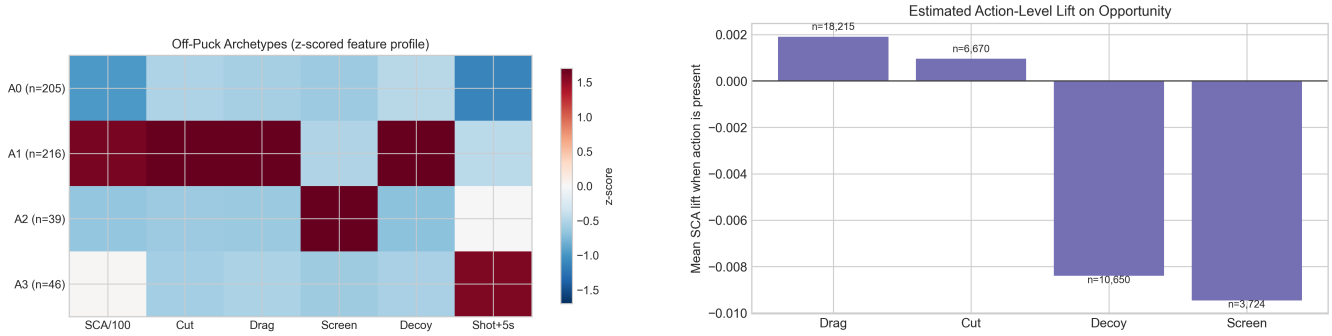

Figure 2: Left: z-scored archetype feature profiles. Right: estimated action lift on frame-level opportunity.

3. Actionable Recommendations (GM / Head Coach)

A. Track SCA as a complementary KPI, not a standalone score. Use SCA alongside zone time and quality-of-opposition controls. Deployment decisions should combine all three.

B. Prioritize drag/cut movement patterns in O-zone structure work. The strongest positive lifts come from lateral manipulation and slot-attacking cuts; build drills around coordinated off-puck lane swaps and timing.

C. Use player archetypes for roster construction and line fit. Balance line trios with at least one high drag/cut manipulator and one stabilizer; avoid stacking low-activity archetypes in top-six usage.

D. Add team-level SCA benchmarking to pre-scout. Current sample suggests strong positive mean possession SCA for Team E / Team I, with negative means for Team L / Team F, indicating different defensive attention burdens to prepare for.

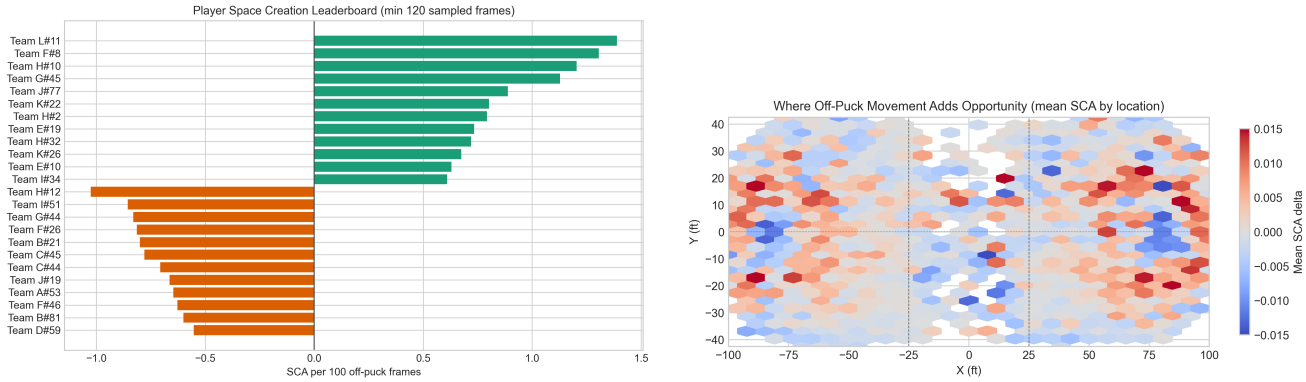

Figure 3: Player-level SCA leaderboard (min sampled frames threshold) and rink map of average SCA contribution by location.

4. Limitations and Next Steps

- Current attribution uses freeze-based approximation, not full Shapley decomposition.

- Action detectors are rule-based and should be calibrated with manual video tags.
- Next revision: include teammate/opponent fixed effects and a stronger counterfactual defense assignment model.

Code + data appendix path: `projects/off_puck_space_creation_value/`